CATALYST. BY MRS. RANA · GRADE 10 · MATH · TERM 1

Term Mock Exam

Built on the ministry exam structure · 21 questions · 100 marks · 150 minutes · calculator allowed

Name _____ Class _____ Score _____

PART 1 · MULTIPLE CHOICE · 17 × 4 MARKS

1 $(6s + 5t) + (4t + 8s) =$ — 4 marks

A $14s + 13t$
B $23st$
C $10s + 13t$
D $14s + 9t$

2 $(x - 2)(x + 6) =$ — 4 marks

A $x^2 - 4x - 12$
B $x^2 + 4x + 12$
C $x^2 - 12$
D $x^2 + 4x - 12$

3 $(2x + 9)^2 =$ — 4 marks

A $2x^2 + 18x + 18$
B $4x^2 + 36x + 81$
C $4x^2 + 81$
D $4x^2 + 18x + 81$

4 Factor $12a^2 + 16a$ completely. — 4 marks

A $4(3a^2 + 4a)$
B $4a(3a + 4)$
C $a(12a + 16)$
D $2a(6a + 8)$

5 Factor $2x^2 - 17x + 21$. — 4 marks

A $(2x - 3)(x - 7)$
B $(2x - 21)(x - 1)$
C $(2x - 7)(x - 3)$
D $(2x + 3)(x + 7)$

6 Factor $81v^2 - 64w^2$. — 4 marks

A $(9v + 8w)^2$
B $(81v + 64w)(v - w)$
C $(9v + 8w)(9v - 8w)$
D $(9v - 8w)^2$

7 The function $f(x) = x^2 + 2x - 6$ has a… — 4 marks

A maximum value of −7
B minimum value of −7
C minimum value of −1
D maximum value of −6

8 Which describes $g(x) = (x - 3)^2 + 1$ compared with $f(x) = x^2$? — 4 marks

A 3 units left and 1 unit up
B 3 units right and 1 unit down
C 3 units right and 1 unit up
D 1 unit right and 3 units up

9 Solve $x^2 + 7x + 14 = 0$ by graphing. (Vertex of the related function: $(-3.5, 1.75)$.) — 4 marks

A $x = -7$ and $x = -2$
B $x = 2$ and $x = 7$
C $x = -3.5$
D no real solutions

10 Solve $4x^2 - 12x = 0$. **4 marks**

A $x = 4$ or $x = 12$
B $x = 0$ or $x = -3$
C $x = 0$ or $x = 3$
D $x = 3$ only

11 Find c so that $x^2 + 26x + c$ is a perfect square trinomial. **4 marks**

A 676
B 13
C 169
D 26

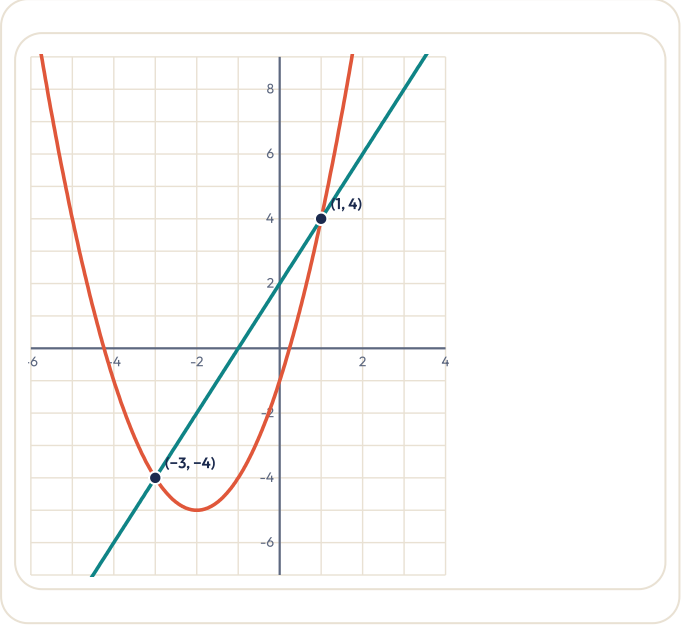
12 Solve $x^2 - 12x = -11$ using the Quadratic Formula. **4 marks**

A −1 and −11
B 1 and 11
C $6 \pm 5i$
D −1 and 11

13 The roots of $x^2 - 5x + 2 = 0$ are... **4 marks**

A two real irrational roots
B no real roots
C two real rational roots
D one real root

14 Use the graph to solve $y = x^2 + 4x - 1$ and $y = 2x + 2$. **4 marks**



A $(-3, -4)$ and $(1, 4)$
B $(0, -1)$ and $(0, 2)$
C $(-4, -3)$ and $(4, 1)$
D $(-1, 0)$ only

15 $(7xy^2)(2x^4y^3) =$ **4 marks**

A $14x^4y^6$
B $14x^5y^5$
C $9x^5y^5$
D $14x^5y^6$

16 Simplify $\frac{a^7b^5c^4}{a^2b^3c}$. **4 marks**

A $a^9b^8c^5$
B $a^{\frac{7}{2}}b^{\frac{5}{3}}c^4$
C $a^5b^2c^4$
D $a^5b^2c^3$

17 Simplify $\frac{a^7b^{-4}}{c^{-2}}$. **4 marks**

A $\frac{a^7}{b^4c^2}$

B $-\frac{a^7c^2}{b^4}$

C $a^7b^4c^2$

D $\frac{a^7c^2}{b^4}$

PART 2 · WRITTEN ANSWERS · 4 × 8 MARKS · SHOW EVERY STEP

21 Use the Distributive Property to find $(2x - 1)(x^2 + 4x - 3)$. Show your steps. **8 marks**

22 Factor $x^2 - 10x + 24$ and $x^2 + x - 20$. Show the factor pairs you tried. **8 marks**

23 Graph $f(x) = x^2 + 2x - 6$. Show the axis of symmetry, the vertex, the y-intercept and a table of values. **8 marks**

24 Solve the system algebraically: $y = x^2 - 6x + 5$ and $x + y = -1$. Check one solution. **8 marks**

Total: _____ / 100

LESS EFFORT. FULL MARK.